

CA#4

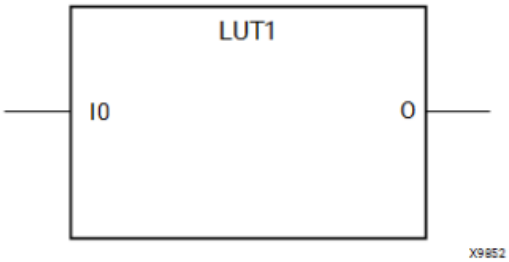
Erfan Falahati - 810102491

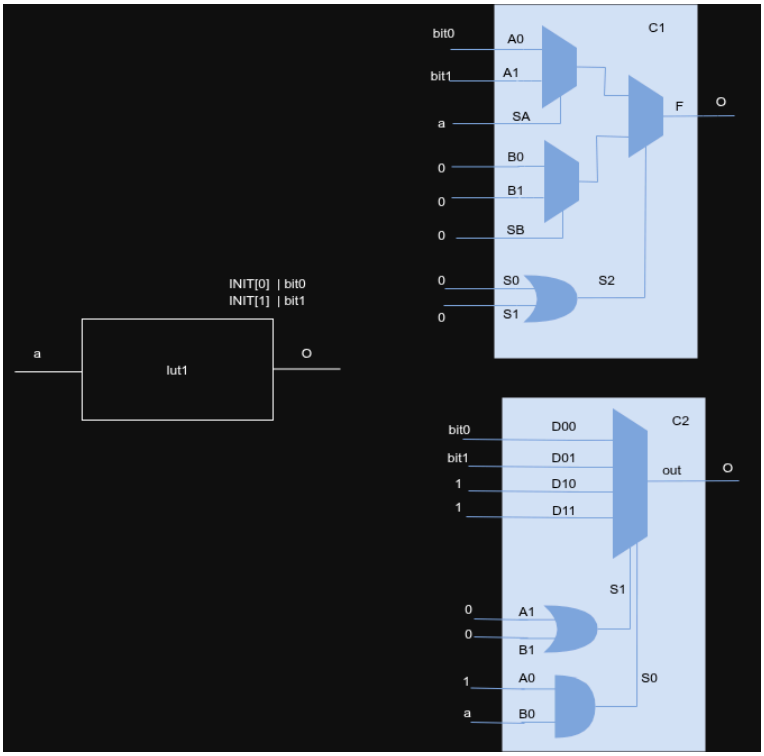
Sadegh Samadi - 810102588

Mapped Blocks

Lut1

Inputs	Outputs
I0	O
0	INIT[0]
1	INIT[1]
INIT = Binary number assigned to the INIT attribute	

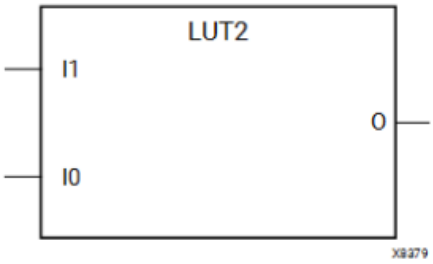


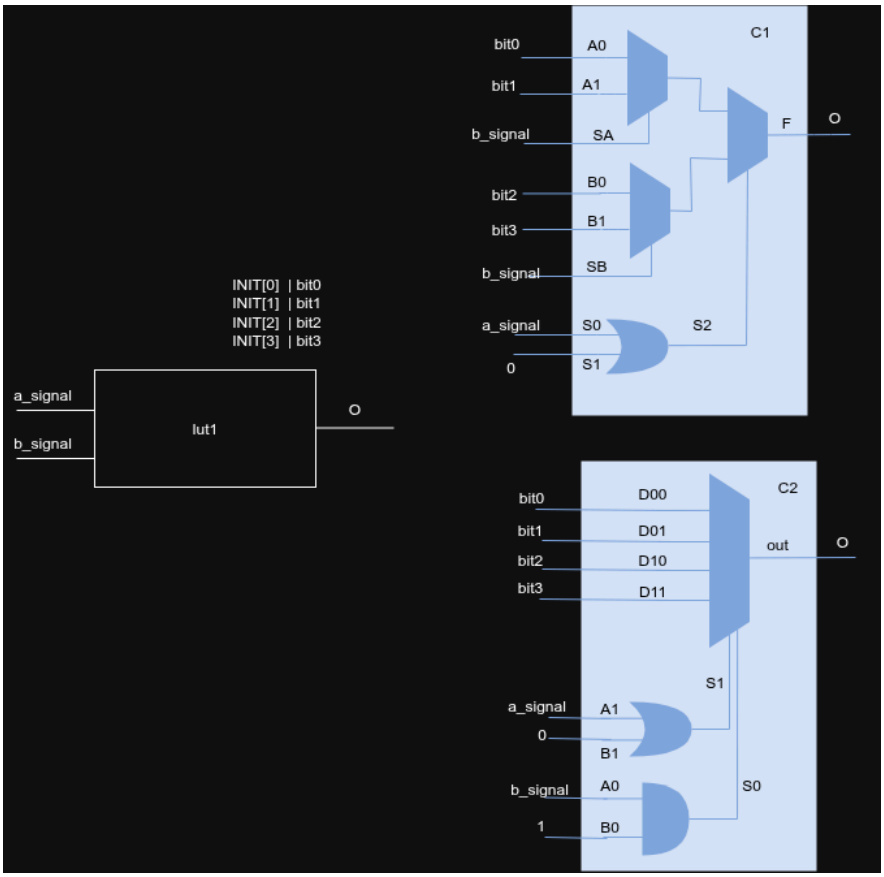


Lut2

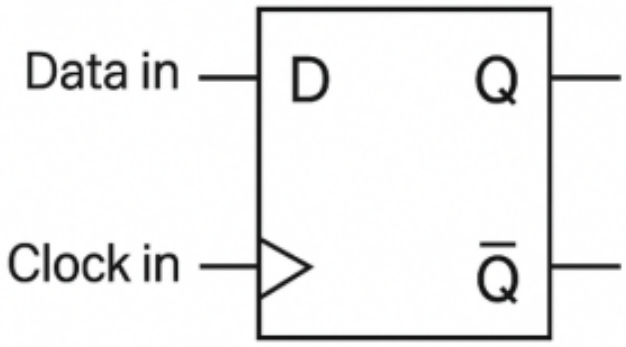
Inputs		Outputs
I1	I0	O
0	0	INIT[0]
0	1	INIT[1]
1	0	INIT[2]
1	1	INIT[3]

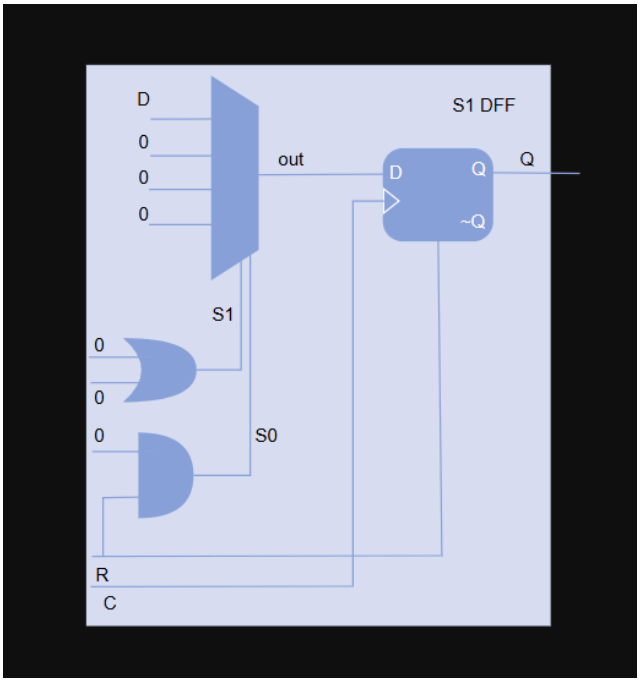
INIT = Binary equivalent of the hexadecimal number assigned to the INIT attribute



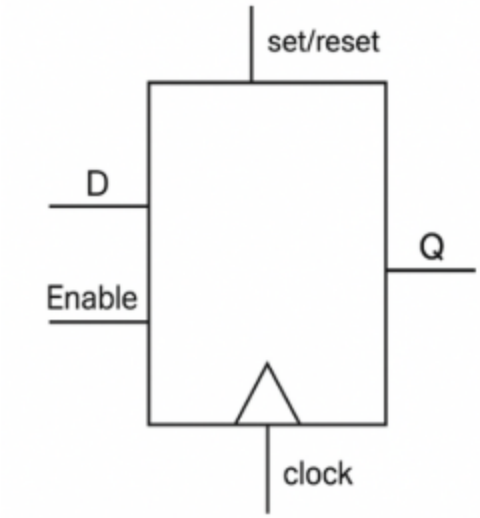


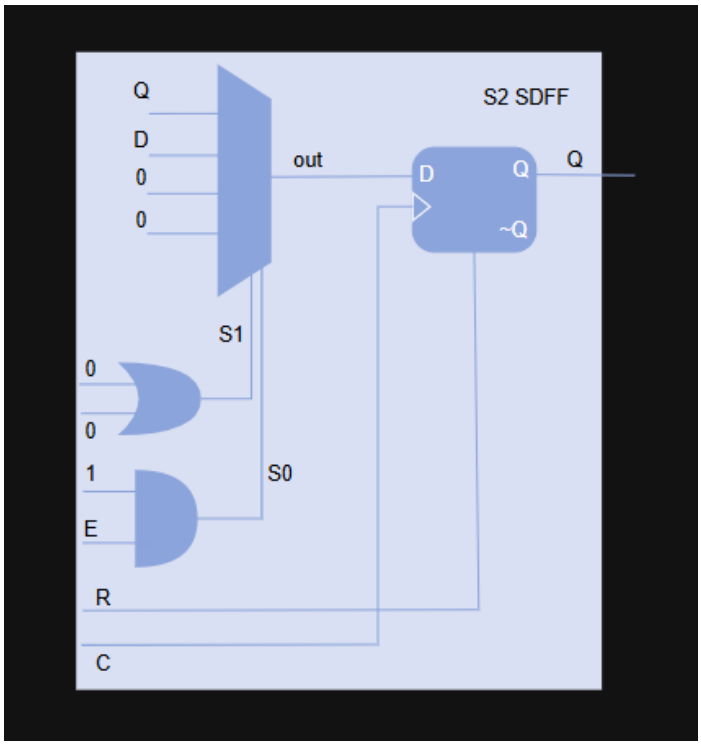
DFF





SDFF





Code synthesis using the Lut1, Lu2, DFF, and SDFF modules with the Yosys tool

```
# Yosys synthesis script for LUT2 mapping
# add your verilog files here like this:
# read_verilog <verilog_file1.v>
# read_verilog <verilog_file2.v>

read_verilog Adder.v
read_verilog controller.v
read_verilog controller2.v
read_verilog counter.v
read_verilog datapath.v
read_verilog datapath2.v
read_verilog functions.v
read_verilog HashGen(Top).v
read_verilog M_file.v
read_verilog mux2.v
read_verilog mux4.v
read_verilog register.v
read_verilog rnd_generator(Top).v
read_verilog shiftRegister.v

hierarchy -check -top HashGenerator
synth -lut 2
write_verilog -norename -noattr -noexpr -nodec mapped_design.v
```

```
erfan@erfan-falahati:~/Desktop/Fall1404/CAD/git/CAD/CA4/code$ yosys lut2synthesis.ys
```

Mapping the gates used by Yosys to Actel blocks using python

```
erfan@erfan-falahati:~/Desktop/Fall1404/CAD/git/CAD/CA4$ python lut2mapper_fixed.py mapped_design.v --cell c1  --count-cells
Found 0 LUT1 and 586 LUT2 instances (out of 586 $lut)
Total number of c1: 586
Total number of S1: 0
Total number of S2: 60
Created mapped design: mapped_mapped_design.v
Created modules file: modules.v
```

TestBench

```
`timescale 1ns/1ns
module HashGenerator_TB();
    reg clk;
    reg rst;
    reg start;
    wire done;
    reg [31:0] message;
    wire [31:0] digest;

    initial clk = 1'b0;
    always #5 clk = ~clk;

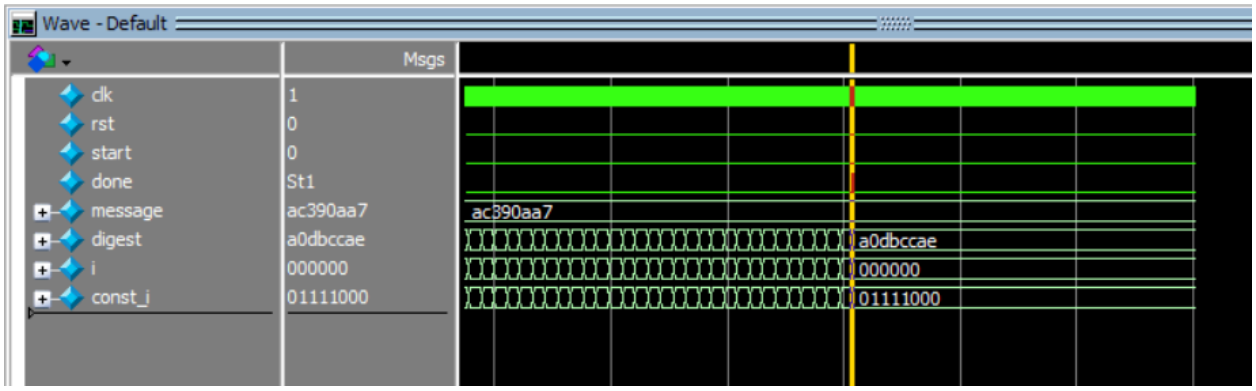
    wire [5:0] i;
    wire [7:0] const_i;

    ConstMem #(
        .INPUT_WIDTH(6),
        .OUTPUT_WIDTH(8),
        .MEM_FILE("constant.mem")
    ) const_mem (
        .index(i),
        .data_out(const_i)
    );

    HashGenerator  hash_generator(
        .clk(clk),
        .rst(rst),
        .start(start),
        .message(message),
        .const_i(const_i),
        .done(done),
        .digest(digest),
        .i(i)
    );

    initial begin
        message = 32'b0; start=0; rst = 1'b1;
        #10 rst = 1'b0;
        #10 start = 1'b1 ; message = 32'hac390aa7;
        #10 start = 1'b0;
        #10000;
        $stop;
    end

endmodule
```



Total number of gates used:

8256

Total number of gates used in CA#3:

3421

3421 / 8256 ≅ 0.44 %